

X-ray topography, photopyroelectric and two-photon absorption studies on solution grown benzimidazole single crystal

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Abstract Benzimidazole (BMZ) is an organic nonlinear optical material, and it is possible to grow it in the laboratory by a slow evaporation solution growth technique using methanol as the solvent. Its nonlinear optical harmonic generation efficiency is 4.5 times higher than that of standard potassium dihydrogen phosphate (KDP) single crystal. It crystallizes with orthorhombic structure, which is noncentrosymmetric in nature, with lattice parameters $a = 13.58 \text{ \AA}$, $b = 6.85 \text{ \AA}$, $c = 6.97 \text{ \AA}$. The functional groups of the grown specimen are confirmed by heteronuclear chemical shift correlation (HETCOR) analysis and it is found that other elements are not incorporated in the compound. The crystalline perfection is examined by high resolution X-ray diffraction (HRXRD), and its topographic image was recorded for the

as-grown single crystal indicates the presence of some grain boundaries in the grown specimen. Thermal parameters such as thermal diffusivity (α), thermal conductivity (k), thermal effusivity (e), and heat capacity (c_p) have been determined by a photopyroelectric thermal wave method. In addition, piezoelectricity, birefringence, and third-order nonlinear optical properties have been examined. The two-photon absorption coefficient of the title compound is determined by the open-aperture Z-scan technique.

1 Introduction

Second- and third-order nonlinear optical materials (NLO) are useful in the area of photonic and optoelectronic devices and technologies [1–3]. In most of these applications, organic, inorganic, and organometallic compounds are widely used to meet the highly demanding technological requirements and specifications. New materials are being grown continuously and their properties investigated with a view to make use of them for NLO applications. During the past few years, several of these materials have shown promise as a highly efficient NLO material, but most of them suffer from practical limitations when compared to a widely used material such as KDP. Research is actively being pursued to identify newer materials, which do not have limitations for practical applications.

In this paper, we report the growth, characterization, and analysis of relevant physical properties of a potential organic nonlinear optical material, Benzimidazole, grown by the solvent evaporation method at room temperature. Benzimidazole (BMZ, with chemical formula $C_7H_6N_2$) or o-phenyleneformamidie is a highly stable heterocyclic molecule, which contains a phenyl ring fused to an imidazole ring. The benzimidazoles have also been called as

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benzimidazoles or benzoglyoxalines. The position of the hydrogen atom attached to the imide nitrogen of the imidazole ring readily changes resulting in the tautomeric form of benzimidazole. This tautomeric character is similar to imidazoles and amidines. Single crystal of the material is non-centrosymmetric in nature. Its structure has been determined and reported by Escande and Galigne [4]. Optical second harmonic properties of BMZ have already been investigated and found that it has higher relative SHG efficiency in comparison with KDP [5]. It crystallizes into the orthorhombic crystal system with the space group $Pna2_1$. We report, for the first time, the structural, topographic, thermal, piezoelectric, heteronuclear chemical shift correlation, and third-order nonlinear optical properties of a solution grown BMZ single crystal. Details of the experimental techniques, results obtained, and a discussion of the results are outlined in the following sections.

2 Experimental techniques

2.1 Crystal growth

The commercially available benzimidazole material was commercially procured and improved its purity by repeated recrystallization processes. The concentrated solution was prepared by using the purified salt with methanol as the solvent. Details of the growth procedure and solubility have already been reported earlier [6]. After a span of about 10 days, good quality single crystals have been harvested, and then used it for characterization analyses.

2.2 Sample characterization

2.2.1 X-ray diffraction

The as-grown single crystal was crushed into fine powder and then subjected to a powder X-ray diffraction measurement with a Bruker D-8 advance X-ray diffractometer at a scan speed of $1^\circ/\text{min}$. The observed two theta values are used as the input for calculating the lattice dimensions. The calculated lattice dimensions are in good agreement with the values reported in literature [4].

The crystalline perfection of the as-grown specimen was examined with a Rigaku double crystal X-ray diffractometer employing a 12 kW Rigaku rotating anode X-ray generator. A (111) Si monochromator of 125 mm diameter was aligned for (333) reflection and highly monochromatic $\text{CuK}\alpha_1$ radiation was obtained. The specimen was used as a second crystal in (+, -) configuration. A good quality BMZ single crystal has been chosen, and its surface was prepared by grinding and hand lapping, and then chemically etched with a water-acetone solution mixture in the ratio of 1:1. BMZ

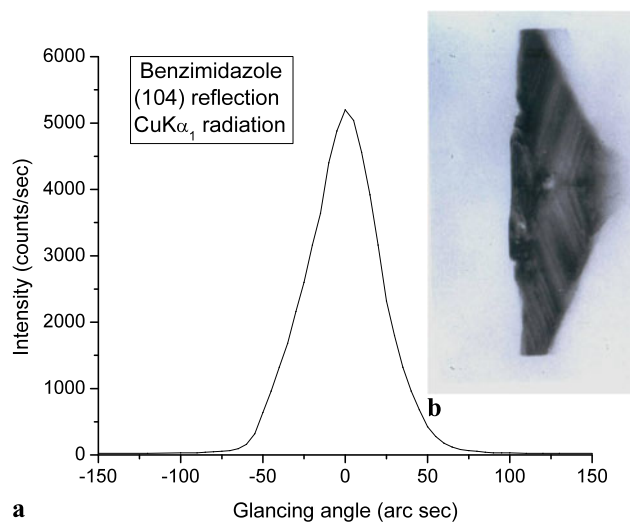


Fig. 1 (a) HRXRD curve of BMZ single crystal and (b) traverse topograph recorded in symmetrical Bragg geometry

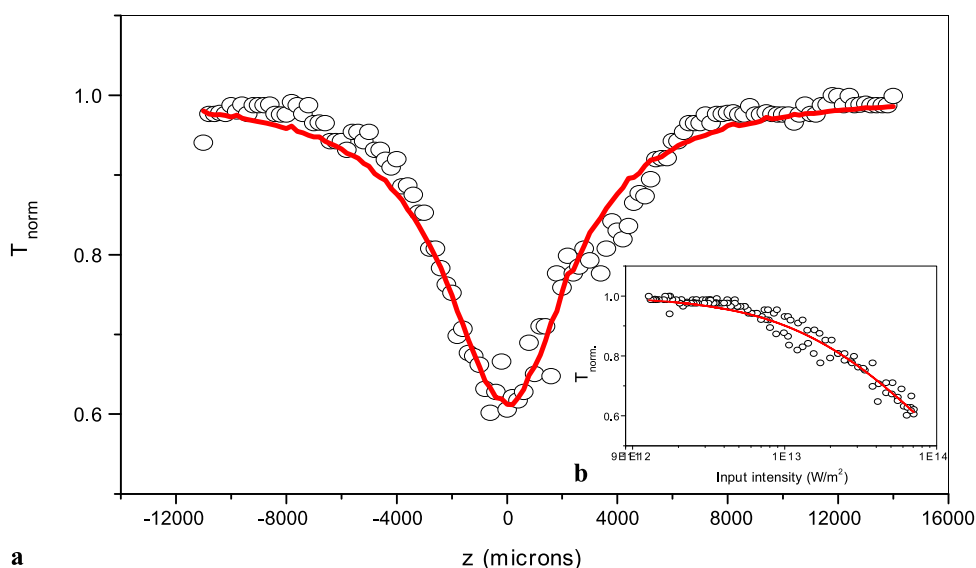
single crystal was aligned for (104) reflecting plane, the corresponding rocking curve (Fig. 1a) and traverse topograph was recorded. The full width at half maxima (FWHM) of this curve is 47 arc sec, which shows that the crystal contains structural grain boundaries. This is normally expected in solution grown single crystals. As a whole, the grown single crystal quality is fairly good.

The traverse topograph has been recorded at the peak position of (104) and it is shown in Fig. 1b. Some growth band can be seen in this topograph, which indicates that during crystal growth, some temperature or super saturation fluctuation has taken place. Also, the intensity is not uniform throughout the topograph, which indicates that the specimen is having some grain boundaries because of the incorporation of solvent molecules [5], which in turn reduces the crystalline perfection of the compound.

2.2.2 Z-scan measurements

The third-order nonlinear optical properties were tested following the well-known open-aperture Z-scan technique [7]. A Q-switched Nd:YAG laser (532 nm, 5 ns pulses) is used as the excitation source. The specimen was well polished and placed on a computer controlled translation stage, which moves the sample along the Z-axis with respect to the beam focus. Pyroelectric energy probes have been used to measure the incident and transmitted energies. The nonlinear absorption coefficient was calculated from the open aperture Z-scan measurement procedure and it is shown in Fig. 2a. As can be seen from the figure, the valley-shaped curve shows the positive nonlinear absorption behavior of BMZ, revealing that the material has an optical limiting action at 532 nm. The optical limiting curve (transmission vs. input intensity) calculated from the Z-scan curve is shown in Fig. 2b. The

Fig. 2 (a) Open aperture Z-scan of BMZ; (b) Optical limiting curve of BMZ



solid lines in the figures are theoretical fits to the experimental data following the nonlinear transmission equation [7]. Two-photon absorption is the mechanism responsible for the observed nonlinear transmission, and the numerically calculated two-photon absorption coefficient of benzimidazole under the present excitation conditions is 3.7×10^{-11} m/W.

2.2.3 Piezoelectric measurement

The piezoelectric response of the benzimidazole single crystal was measured using a piezometer system (PM200) at 110 Hz employing a dynamic force of 0.25 N at ambient temperature. The cut and well polished surface of the specimen was chosen and then coated with a silver paste uniformly on both the surfaces for making good electrical contacts (thickness ~ 0.8 mm). Then the sample was carefully loaded in between the perpendicular electrodes and then the force was applied gently. From this method, we obtained the d_{33} and d_{31} coefficients as 0.1 pC/N and 1.85 pC/N, respectively.

2.2.4 Thermal properties by photopyroelectric technique

In the recent past, the photopyroelectric (PPE) thermal wave technique has gained popularity as a convenient technique to determine thermal properties of delicate materials such as single crystals. The technique enables one to measure different thermal transport properties of single crystals thermal diffusivity (α), thermal conductivity (k), thermal effusivity (e), and heat capacity (c_p) [8]. These parameters for a nonlinear optical crystal are critical for its use to fabricate devices because the optical damage threshold of a crystal is determined by its thermal conductivity and specific heat capacity, which determine the heat dissipation capability of the material. So, a measurement of the thermal properties of

an NLO material is highly significant. We have carried out these measurements on a good quality, optically polished BMZ specimen following the procedure described in earlier reports [8]. Measurement of the PPE signal phase and amplitude enables one to determine the thermal diffusivity (α) and thermal effusivity (e). From the measured values of α and e , thermal conductivity (k) and heat capacity (c_p) of BMZ were calculated and tabulated (Table 1).

2.2.5 Birefringence

The well polished wafer of BMZ specimen was subjected to birefringence measurements in an interferometer with a laser wavelength of $\lambda = 6328$ Å for measuring the variation in birefringence of the crystal along the thickness of the sample ($L = 0.39$ mm), which is along the [201] crystallographic plane. The details of the instrumentation for this measurement have already been reported [9, 10]. A birefringence interferogram obtained for our crystal plate is shown in Fig. 3. The variation in birefringence, which is calculated along the c axis of the crystal is obtained as $\Delta(n_e - n_o) = 0.00162256$. The uniformity of fringes signifies the optical homogeneity of the sample, and does not contain any volume defects such as precipitates in the sample. Moreover, no lattice misorientations due to local sharp changes in the dislocation density exist across the sample.

2.2.6 HETCOR analysis

The ^1H , ^{13}C NMR and heteronuclear chemical shift correlation (HETCOR) experiment was carried out to get a better insight in to the composition of the crystals. The HETCOR experiment is an important tool to distinguish between the protons and carbons present in the molecule and to describe their environment. The ^1H NMR spectrum of BMZ

Table 1 Thermodynamical parameters of BMZ

Sample	Density, ρ (kg/m ³)	Thermal effusivity e (Ws ^{1/2} /m ² K)	Thermal diffusivity α ($\times 10^{-6}$ m ² /s)	Thermal conductivity k (W/mK)	Sp. heat capacity c_p (J/kg K)
BMZ	1230	3250 $\pm$ 6	2.40 $\pm$ 0.61	5.00 $\pm$ 0.44	1746 $\pm$ 22

Fig. 3 Recorded birefringence pattern of BMZ

(300 MHz, DMSO- d_6) revealed three singlet at $\delta_H = 7.25$, 7.70, and 8.30 ppm in the aryl region. The ^{13}C NMR revealed peaks at 116.0, 122.0, and 142.1 ppm for the carbon nuclei, which contain protons whereas a small peak appeared at 138.3 ppm for carbon which contain no protons.

3 Conclusions

A single crystal of the potential nonlinear optical material benzimidazole has been successfully grown by the slow evaporation solution growth technique at room temperature using methanol as the solvent. Powder X-ray diffraction studies reveal that the crystal belongs to the orthorhombic crystal system and HETCOR analysis confirms the presence of functional groups of the title compound. The crystalline perfection has been examined by high resolution X-ray diffraction and topography measurements. The piezoelectric coefficient and thermal properties have been eval-

uated. From open aperture, Z-scan measurements using 5 ns laser pulses, the two-photon absorption coefficient at 532 nm is calculated to be 3.7×10^{-11} m/W under the present excitation conditions.

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